

Total No. of Questions : 6]

SEAT No. :

P4

[Total No. of Pages : 1

APR-17/BE/Insem.-4

B.E. (Civil)

ADVANCED FOUNDATION ENGINEERING

(w.e.f. - June 2015) (2012 Pattern) (Elective - III(2))

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6.*
- 2) *Figures to the right indicate full marks.*

- Q1)** a) Explain IRC provisions for subsoil exploration of roads. [5]
b) Explain any one case study of failure of deep foundation with proper justification of failure. [5]

OR

- Q2)** a) Explain depth of soil exploration for different types of foundations as per IS 1892. [5]
b) Explain in detail the soil exploration program for construction of a bridge. [5]

- Q3)** a) Explain uplift piles and Anchor piles with suitable sketches. [5]
b) Explain cyclic load test for piles with suitable sketch. [5]

OR

- Q4)** a) Write a short note on racer piles. [5]
b) What are the I.S. code provisions for design of deep foundation. [5]

- Q5)** a) Draw a neat sketch of under reamed pile foundation & explain functions of each component parts. [5]
b) Explain in detail Design considerations for sand drains. [5]

OR

- Q6)** a) Write a static formula to estimate load carrying capacity of under reamed piles & explain meaning of each term. [5]
b) Write a short note on stone columns. [5]

